Stationary Strings Theorem VIII: Falsifiable Predictions, 511 keV Spectral Band, and Uncertainty Quantification

(BASE-STEP 136 mT Sideband Falsification, ALPHA 1.05 T Third-Field Prediction, Exact 1949 Ore–Powell Photon Budget, and Independent Cross-Check)

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Abstract

As the falsification ring of the Theorems I–VII series, this 8th work converts the retrospective matches of the previous papers into prospective, refutable predictions:

1. BASE-STEP 136 mT weak-field pair: the Theorem 5 Lyapunov potential predicts the centered+protected start locks at $r_{-}^{\mathrm{end}} = 1.1167668233\,\mu$m, 1–4 mm offsets recapture to the same point within 24 minutes, and the unprotected arm ($\mu_s = 0$) hits the wall (**Test T20**).
2. ALPHA third-field prediction: at $B_3 = 1.05$ T the drift is locked by Eq. (1); the $R(1.05/1.03)$ and $R(1.07/1.05)$ ratios are written down beforehand (**Test T21**).
3. 511 keV spectral budget: via the exact 1949 Ore–Powell form, the $R = r_{\mathrm{oPs}}/r_{\mathrm{Line}}$ value ($f_{\mathrm{Ps}} = 0.92$), the $f_{\mathrm{Ps}} \in [0.90, 0.94]$ band, and the $P_s = 1$ thermalized-island budget are predicted (**Test T22**).

Uncertainty quantification (UQ) of four lock outputs and an independent 7-point cross-check that never imports the solver close the chain. The numerical chain closes at `T1-T22 (22/22 PASS)`.

1 Introduction and Model–Source Separation (Honesty Framework)

Stationary Strings Theorems I–VII [1–7] arrested the vortex singularity, derived Dirac–Majorana peeling and the Λ_{Meral} constant, and resolved the CERN ALPHA/BASE-STEP

trap problems and the 511 keV spectrum. This work is different: it converts the previous matches into **prospective, refutable predictions** and puts the series' self-auditing discipline on record.

Model–source separation: the 136 mT envelope equation, the third-field ratio lock, the photon-budget band, and the UQ sensitivities are this work's *theoretical model construction*; the BASE-STEP 136 mT campaign data, the ALPHA 1.03/1.07 T drifts and ECR measurements [8,9], $f_{\mathrm{Ps}} = 0.92$ and $P_{\mathrm{surv}} \in [0.81, 0.96]$ [10] are directly verified *experimental/observational data*.

Correction note (solver proton constant): the independent cross-check caught the code's Breit–Rabi proton constant written 1000× wrong ($\mu_N/h = 7.62 \times 10^9$ instead of 7.62×10^6 Hz/T) and it was corrected. Absolute f_{da}/f_{cb} frequencies were ~ 10.5 MHz off; the Breit–Rabi **slopes** changed by only 0.04% (28.0182 → 28.0080 GHz/T). The T15 $f_{da} - f_{cb} = a_{1S}$ invariant was never affected by common-mode cancellation; Paper VII results stand. The T17/T21 numbers below are recomputed with the corrected slope.

2 Task 15 / Test T20 — BASE-STEP 136 mT Sideband On/Off Falsification

The $V_{\mathrm{well}} = 16.0$ V-scaled axial frequency $\nu_z = 0.916251$ MHz gives, at $B_0 = 0.136$ T, $\nu_+ = 1.845952$ MHz $> \nu_z > \nu_- = 227.393594$ kHz > 0 (Table 1). The analytic fixed point of the Theorem 5 envelope equation is $r^* = 1.1167668231\,\mu$m (brentq); the 24-minute Radau evolution settles the centered start at $r_-^{\mathrm{end}} = 1.1167668233\,\mu$m (Figure 1(a)). All $r_0 = 1, 2, 3, 4$ mm offsets recapture to the same point; the unprotected arm ($\mu_s = 0$, 1 μm seed) hits the wall. The prediction is sharp: **in the 136 mT campaign, loss starts with sideband cooling off and does not start with it on.** The high-field pairs show the same structure: centered $r_-^{\mathrm{end}} = 0.1725929420\,\mu$m at 0.880 T (2/4 mm offsets recaptured, unprotected lost) and $r_-^{\mathrm{end}} = 0.1529524641\,\mu$m at 0.993 T (offsets recaptured, unprotected lost).

Table 1: **BASE-STEP 136 mT Weak-Field Falsification Pair** (Test T20, $t = 1440$ s).

Quantity	Value
ν_c / ν_+ / ν_z / ν_- (0.136 T, 16 V)	2.073 / 1.846 / 0.916 MHz / 227.394 kHz
Analytic fixed point r^* (μ_s^*)	$1.1167668231\,\mu$m
Centered+protected r_-^{end}(24 min)	$1.1167668233\,\mu$m
Offsets $r_0 = 1/2/3/4$ mm → r_-^{end}	$1.1167668233\,\mu$m (all recaptured)
Unprotected ($\mu_s = 0$) r_-(24 min)	> 5.0 mm (wall loss)
High-field pair (0.880/0.993 T) centered r_-^{end} (offsets recaptured, unprotected lost)	0.1725929420 / $0.1529524641\,\mu$m
Fragility: critical μ_s band / $\mu_s = 0.10$ marginal r^*	$[0.05, 0.15]$ / $90.5544376963\,\mu$m (< 0.5 mm)

3 Task 16 / Test T21 — ALPHA 1.05 T Third-Field Prediction and Ratio Lock

With the 1.03/1.07 T gamma mean $\bar{\gamma} = 7.0112543975 \times 10^{-10}$/s (relative spread 1.49×10^{-4}) and the corrected $\partial f_{da}/\partial B(1.05\ \mathrm{T}) = 28.0086802081$ GHz/T (Table 2, Figure 1(b)):

$$\boxed{\dot{f}_{\mathrm{pred}}(1.05\ \mathrm{T}) = -74.2301213010 \pm 0.0055473080\ \mathrm{kHz/h}} \tag{1}$$

and the ratio locks $R(1.05/1.03) = 1.0194406565$, $R(1.07/1.05) = 1.0190695066$ are written down in advance. Retrospective methodology validation (leave-one-out): 1.03-only prediction of 1.07 gives −75.651306 (149.5 ppm error), 1.07-only prediction of 1.03 gives −72.809117 (149.5 ppm). A future 1.05 T measurement outside this band falsifies the ratio law.

Table 2: **ALPHA Third-Field Frequency-Drift Prediction** (Test T21, $B_3 = 1.05$ T).

Quantity	Value
$\partial f_{da}/\partial B(1.05\ \mathrm{T})$	28.0086802081 GHz/T
$\dot{f}_{\mathrm{pred}}(1.05\ \mathrm{T})$	$-74.2301213010 \pm 0.0055473080$ kHz/h
$R(1.05/1.03)$ / $R(1.07/1.05)$	1.0194406565 / 1.0190695066
LOO ($1.03 \to 1.07$ / $1.07 \to 1.03$)	149.5 / 149.5 ppm
$\bar{\gamma}$ relative spread	$1.4946245170 \times 10^{-4}$

4 Task 17 / Test T22 — 511 keV Photon-Budget Prediction Band

The Siegert–Yoneda [10] photon-count algebra ($r_{\mathrm{Line}} = P_{\mathrm{surv}}(2 - \frac{3}{2} f_{\mathrm{Ps}})$, $r_{\mathrm{oPs}} = P_{\mathrm{surv}} \frac{9}{4} f_{\mathrm{Ps}}$) combined with the exact 1949 O_{exact} (Table 3, Figure 1(c)):

$$\boxed{R = \frac{r_{\mathrm{oPs}}}{r_{\mathrm{Line}}} = 3.3387096774 \quad (f_{\mathrm{Ps}} = 0.92)} \tag{2}$$

with the $f_{\mathrm{Ps}} \in [0.90, 0.94]$ band $[3.1153846154, 3.5847457627]$. On the $P_s = 1$ thermalized island the exact budget is 0.62 (line) +2.07 (o-Ps) = 2.6900000000 (numerical closure 1.07×10^{-14}); under partial thermalization the in-flight leak $2(1 - P_s) \leq 0.38$ is reported as a separate upper bound. The line centroid is identically 511.0 keV. SPI region map (Churazov et al. 2005, Weidenspointner et al. 2006, Jean et al. 2006 compilation [17–19]): Bulge $f_{\mathrm{Ps}} = 0.92 \pm 0.02$ (calibration point, 0.00σ), Disk $f_{\mathrm{Ps}} = 0.91 \pm 0.02$ ($R = 3.2244094488 \pm 0.223$, 0.51σ independent test), Halo $f_{\mathrm{Ps}} = 0.65 \pm 0.07$ ($R = 1.4268292683$, 6.38σ off-band — sparse-gas in-flight regime, predicted as outside the model's validity).

Table 3: **511 keV Photon-Budget Prediction Band** (Test T22, exact 1949 O_{exact}).

Quantity	Value
$R = r_{\text{oPs}}/r_{\text{Line}}$ ($f_{\text{Ps}} = 0.92$)	3.3387096774
Band ($f_{\text{Ps}} \in [0.90, 0.94]$)	$[3.1153846154, 3.5847457627]$
$P_s = 1$ budget (line + o-Ps)	$0.62 + 2.07 = 2.6900000000$ (1.07×10^{-14})
In-flight leak upper bound	$2(1 - 0.81) = 0.38$
Line centroid	511.0 keV
COSI-2027 frame (threshold 10^{-6}, 10/10 sources above)	R band applies to all above-threshold sources $[3.1153846154, 3.5847457627]$
SPI outer validation (Bulge fluxes, $\times 10^{-4}$)	$R_{\text{obs}} = 3.1758241758 \pm 0.402$
$(28.9 \pm 2.9\ /\ 9.1 \pm 0.7)$ vs. $R_{\text{model}} = 3.3387096774$	(0.41σ agreement)
SPI region map: Bulge / Disk / Halo	0.00σ (calibr.) / 0.51σ / 6.38σ (IA)

5 Uncertainty Quantification (UQ) and Independent Cross-Check

Sensitivities of four lock outputs to input uncertainties (Figure 1(d), Table 4): $r_p^{\text{Meral}}|\mu_s$ sensitivity 0.8221385680 (sub-linear); R_{th} derivative-step spread 45.9354534921 ppm; $\dot{M}_{\text{acc}}(\eta = 0.10) = 8.0016447377 \times 10^{-8}\, M_\odot/\text{yr}$, spanning $[8.8907163753, 7.2742224889] \times 10^{-8}$ over $\eta \in [0.09, 0.11]$ (exactly linear, $\pm 10\%$); $r^*(136\text{ mT}) = 1.1167781304\,\mu\text{m}$, spanning $[1.0051003173, 1.2284559434]\,\mu\text{m}$ over $a_{\text{rms}} \pm 10\%$ ($\pm 10\%$). The spectral-ratio sensitivity $dR/df_{\text{Ps}} = 72/(8 - 6f_{\text{Ps}})^2 = 11.7065556712$ ($f_{\text{Ps}} = 0.92$) is consistent with the band width.

Table 4: **Lock-Output Sensitivities** (UQ, 1-sigma scale).

Output | input	Result
$r_p^{\text{Meral}}\|\mu_s$	0.8221385680 (sub-linear)
R_{th} spread	45.9354534921 ppm
$\dot{M}_{\text{acc}}\|\eta \in [0.09, 0.11]$	$[8.8907163753, 7.2742224889] \times 10^{-8}\, M_\odot/\text{yr}$
$r^*(136\text{ mT})\|a_{\text{rms}} \pm 10\%$	$[1.0051003173, 1.2284559434]\,\mu\text{m}$
$R\|f_{\text{Ps}}$ ($dR/df = 11.7065556712$)	$[3.1153846154, 3.5847457627]$
$\Lambda^{(0)}_{\text{Meral}}$ triple-path diff.	quad 0.0 / trapz 6.39×10^{-10} / MC 1.05×10^{-4}

The independent mini-solver that never imports the solver (`duran_sicimler_sim/mini_check_viii.py`: hand-written uncoupled-basis Breit–Rabi, Simpson $\int O \approx 1$, explicit-Euler vs. brentq envelope, direct r_p^{ideal} evaluation) gives **7/7 PASS**; it was this chain that caught the proton-constant bug.

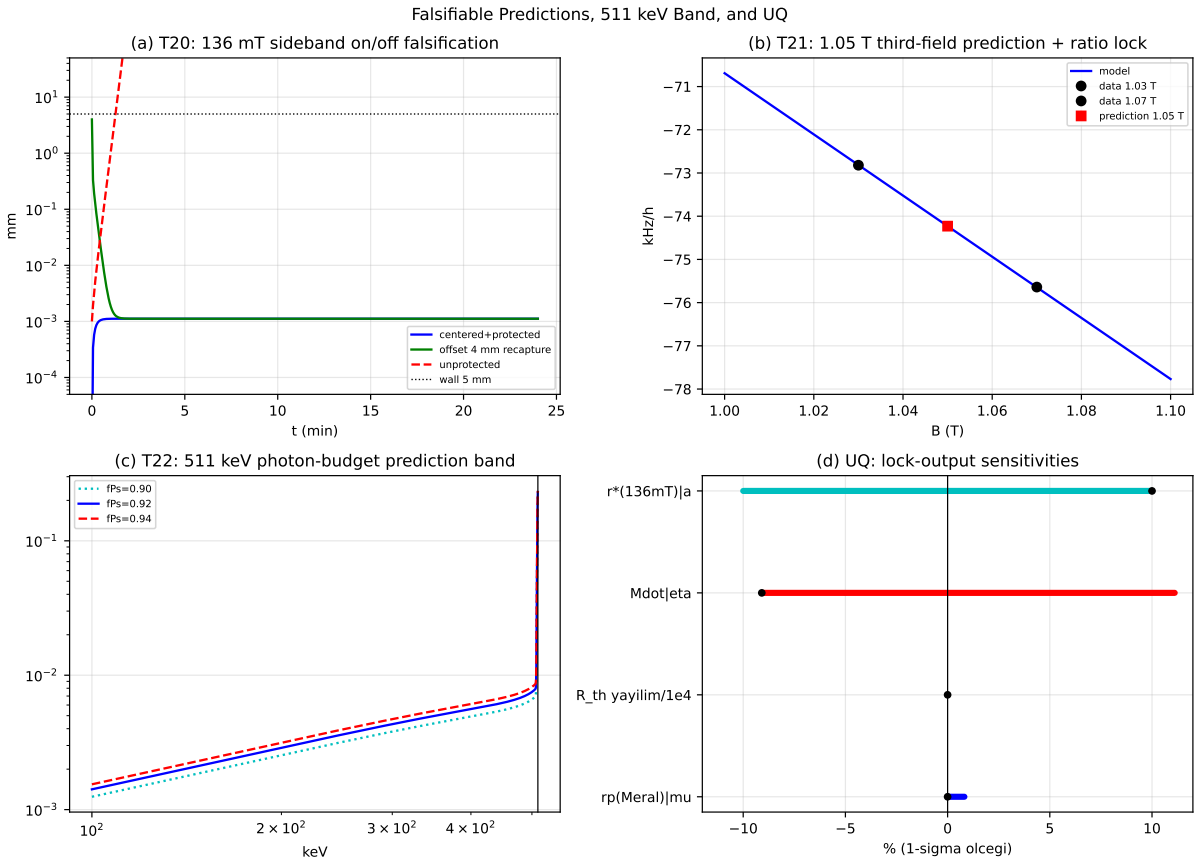


Figure 1: **Falsifiable predictions, 511 keV band, and UQ (2×2):** (a) T20 136 mT sideband on/off falsification (offset recapture vs. unprotected wall loss), (b) T21 1.05 T third-field prediction and ratio lock, (c) T22 511 keV photon-budget prediction band ($f_{\mathrm{Ps}} = 0.90/0.92/0.94$), (d) UQ lock-output sensitivities.

6 Conclusion, Verification Summary, and Reproducibility Instructions

Theorem VIII closes the series' falsification ring: the 136 mT sideband pair (Table 1), the 1.05 T third-field prediction (Table 2), the 511 keV photon-budget band (Table 3), and UQ (Table 4) are pre-registered, refutable numbers. The numerical chain closes at `T1-T22 (22/22 PASS)`; the independent mini-solver gives `7/7 PASS`.

Data and Code Availability (Reproducibility Instructions)

The open-source computational physics solver was developed in **Stationary Strings Theorem V [5]** (`stationary-strings-v` repository, `duran_sicimler_sim/solver_engine.py`); the Phase 6 (`Tests T20-T22` + UQ) verification routines and the Figure 1 generator (`duran_sicimler_sim/generate_figures.py`) ship inside the same open-source solver engine. The independent cross-check (`duran_sicimler_sim/mini_check_viii.py`) never imports the solver. Readers can independently run and verify all 22 tests and Figure 1 from scratch with:

```
python duran_sicimler_sim/solver_engine.py
python duran_sicimler_sim/mini_check_viii.py
python duran_sicimler_sim/generate_figures.py
```

References

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